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Towards Realistic Ultrasound Phantoms of the Mitral Valve

Thickness Reconstruction and Manufacturing of Leaflet Structures from Volumetric Ultrasound Data

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Declaration of Authorship

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I hereby declare that I have written this thesis independently, that I have completely specified the utilized sources and resources and that I have definitely marked all parts of the work – including tables, maps and figures – which belong to other works or to the internet, literally or extracted, by referencing the source as borrowed.
I further declare that I have used generative AI tools only as an aid, and that my own intellectual and creative efforts predominate in this work. In the appendix “Overview of Generative AI Tools Used” I have listed all generative AI tools that were used in the creation of this work, and indicated where in the work they were used. If whole passages of text were used without substantial changes, I have indicated the input (prompts) I formulated and the IT application used with its product name and version number/date.

Vienna, September 28, 2026

Sebastian Bolek

Acknowledgements

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Abstract

The development of realistic, patient-specific cardiovascular phantoms requires accurate anatomical reconstruction and manufacturing methods that preserve the geometric properties of the structures being modelled. In particular, the spatial distribution of mitral valve leaflet thickness represents an important aspect of producing anatomically representative physical models.

This thesis presents a reconstruction and manufacturing pipeline for producing a three-dimensional printed mitral valve prototype with spatially varying leaflet thickness, derived from volumetric ultrasound data. The approach combines a volumetric ultrasound dataset with initially unthickened leaflet surface meshes. After aligning both datasets within a common coordinate system, leaflet thickness is recovered through manually guided measurements at locations where the tissue boundaries can be reliably identified. To account for image noise and regions with insufficient visibility, representative median values are assigned to the central and outer rim regions. The resulting measurements are distributed across the leaflet surfaces using inverse distance weighting. The interpolated thickness distributions are subsequently transferred to the original meshes and converted into closed solid geometries.

The reconstructed model is manufactured using fused deposition modeling with flexible thermoplastic polyurethane filament. Several manufacturing iterations are conducted to address challenges related to material flexibility, support structures, and print quality. The resulting physical prototype is subsequently evaluated by Siemens Healthineers using ultrasound imaging. Feedback from clinical personnel confirms that the printed valve is visible and scannable using ultrasound equipment.

The results demonstrate the feasibility of reconstructing spatially varying leaflet thickness from volumetric ultrasound data and transferring the resulting geometry into a physical, ultrasound-visible prototype. While the initial ultrasound assessment is encouraging, further quantitative evaluation is required to establish acoustic similarity to biological tissue and validate the reconstructed thickness distribution. The developed pipeline provides a foundation for the further development of patient-specific cardiovascular ultrasound phantoms.

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CHAPTER 1

Introduction

This work was motivated by the need for anatomically accurate, patient-specific heart valve phantoms for pre-procedural planning and device testing. The following sections outline the motivation, aim, research question, and scientific contribution of this thesis. The data which this work heavily relies on stems from Siemens Healthcare Diagnostics GmbH (*Siemens Healthineers*), who provided the volumetric ultrasound data and the unthickened surface mesh of the mitral valve leaflets. The pseudoanonymised imaging data was acquired from a patient with mitral regurgitation. The malfunction in the valve was deliberately kept within the data to ensure that the resulting phantom is clinically relevant and can be used, i.e., for flow simulations of a patient specific diseased mitral valve.

Besides the ultrasound data of the valves Siemens Healthineers also provided the imaging data of the atria and ventricles of the heart from the same patient. However, since the substance of producing a physical phantom prototype of the mitral valve leaflets is already complex enough, this work focuses on the leaflets only. A similar task of producing a physical phantom of the atria and ventricles is done by a colleague and fellow student *Peter Machoritsch* at the Technical University of Vienna in parallel to this work. The two works are independent of each other, however some parts especially in Section 3 may overlap as we collaborated during our theses. At the end, the resulting physical phantoms should be combinable to form a complete, watertight heart phantom which is the goal of the overall project. It should be emphasized that whenever certain design-choices with respect to each other's works are made, they are documented and justified to ensure reproducibility and scientific rigor.

1.1 Motivation

Valvular heart disease represents a major and growing global health burden. Among non-rheumatic valve diseases, aortic stenosis and degenerative mitral valve disease are particularly prevalent in ageing populations [CSW+24]. Echocardiography, commonly referred to as cardiac ultrasound, is the principal imaging method for evaluating valvular anatomy, leaflet motion, and blood flow behavior (*haemodynamics*). TransThoracic Echocardiography (*TTE*) is generally used as the non-invasive first-line examination, whereas TransEsophageal Echocardiography (*TEE*) is a semi-invasive technique used when TTE is inconclusive or more detailed visualization is required [PBL+25]. IntraCardiac Echocardiography (*ICE*) uses a catheter-mounted ultrasound transducer introduced into the cardiac chambers and is primarily employed for real-time guidance during catheter-based interventions [ESR+18]. However, echocardiographic image acquisition and interpretation are operator-dependent and may be affected by intra- and inter-observer variability, imaging artifacts, and, particularly for TTE, limited acoustic windows [RRO+20].

To improve diagnostic accuracy, train clinicians, and benchmark automated quantification algorithms, physical cardiac phantoms are increasingly utilized. Traditionally, these phantoms serve as standardized test objects for flow measurement and leaflet segmentation. Moreover patient-specific phantoms are also widely used for planning and simulating cardiovascular interventions [IBN+22]. For complex structural heart interventions, such as Transcatheter Mitral Valve Replacement (*TMVR*) [IOI+16] or Transcatheter Edge-to-Edge Repair (*TEER*), anatomically accurate physical phantoms enable pre-procedural fit assessment, mechanical tissue-device interaction testing, and post-implantation flow simulation [VPJGAL17].

In this context, the geometric accuracy of the phantom — specifically the spatial distribution of leaflet thickness — plays a crucial role. Heart valve leaflets are not uniform surfaces; their thickness varies continuously across different anatomical zones (e.g., from the thin leaflet center to the thicker outer rim) [RFS+13]. This variation affects tensile, bending, and buckling behavior; consequently, assuming a uniform leaflet thickness can introduce substantial errors in the predicted mechanical response [LMM+23]. Creating an anatomically precise, patient-specific, thickness-accurate phantom directly from clinical volumetric ultrasound data remains a technical challenge due to speckle noise, low contrast-to-noise ratios, and spatial resolution constraints.

1.2 Aim of the Work

The primary aim of this work is to design and a pipeline that produces a 3D-printed prototype of an anatomically accurate mitral valve ultrasound phantom with realistic, spatially varying leaflet thickness. Ideally this component would be integrated into a larger heart phantom and allow for a realistic blood-flow simulation. Starting from a volumetric ultrasound scan and an unthickened single-surface leaflet mesh, this project establishes an end-to-end processing pipeline to recover local leaflet thickness from image data, transfer it onto the geometric mesh, construct a closed solid model, and manufacture a flexible physical phantom suitable for ultrasound imaging and device interaction assessment.

1.3 Research Question

To guide the development and evaluation of the pipeline, this thesis addresses the following central research question:

> *To what extent can spatially varying leaflet thickness be recovered from volumetric ultrasound data and transferred to a 3D-printable, anatomically accurate valve model?*

1.4 Scientific Contribution

The key contributions of this work are:

1. **Hand-guided, locally validated thickness recovery:** A hybrid measurement strategy that avoids the pitfalls of global volumetric thresholding in noisy ultrasound data. This is done by combining local intensity profile criteria with guided manual sampling in reliable image regions, supplementing missing values by robust median estimates in low-quality regions.

2. **Multi-software bounding-box alignment:** A reproducible geometric registration method using bounding-box reference planes. This is employed to align disparate coordinate spaces across MeVisLab, 3D Slicer, and Blender without relying on ambiguous curved and complex surface landmarks.

3. **Spatially interpolated boundary surface extrusion:** A method to map discrete thickness measurements across complex surface meshes using inverse distance weighting (IDW) and single-sided boundary surface extrusion. This supports preserving authentic anatomical thickness gradients without artificial uniform wall assumptions.

CHAPTER 2

Related Work

Research related to the present work spans several domains, as the development of a realistic cardiovascular phantom involves anatomical reconstruction, fluid dynamics, physical simulation, and manufacturing. From a fluid-dynamic perspective, it is necessary to first consider what can already be achieved using purely computational simulations. Such models make it possible to study cardiac motion, blood flow, and the interaction between different parts of the cardiovascular system under controlled conditions. At the same time, their results depend on assumptions and simplifications concerning the geometry, material properties, and physiological behavior of the modelled structures [BRF$^+$14, SWH$^+$12, FPR$^+$23].

In contrast, physical cardiovascular simulators and phantoms provide an experimental environment in which blood flow, imaging methods, and medical devices can be examined directly. This raises the question of which advantages a physical reconstruction offers compared with a purely computational model and how such a reconstruction can be adapted to the anatomy of an individual patient. Patient-specific models require suitable methods for extracting anatomical information from medical image data and converting it into a manufacturable geometry. Several approaches for producing such prototypes have already been presented, including additive manufacturing and mould-based fabrication techniques [SPF$^+$24, ESP$^+$19].

This chapter therefore reviews computational cardiac simulations, physical cardiovascular simulators, patient-specific phantoms, and the manufacturing methods used to produce them. Particular attention is given to approaches relevant to the reconstruction and fabrication of anatomically accurate heart-valve structures.

2.1 Computational Cardiovascular System Simulation

Using computational models to simulate the cardiovascular system can be used to study how the heart performs under healthy as well as diseased conditions. The benefit of adapting these methods instead of reverting to experiments involving patients or animals is for one the ethical aspect of completely eliminating the need for living subjects but also, when not relying on artificial physical replicas, a way to study the cardiovascular system in a highly controllable and reproducible environment [PBL$^+$17]. The level of detail can hereby vary widely. While some models this section will cover shift their attention to the cellular level, others describe the movement of the ventricles, the flow of blood, or even the behavior of the entire heart. Combining these different levels is challenging because the electrical activation of the heart, the contraction of the muscles, and the resulting blood flow all influence each other. The *University of Tokyo heart simulator* is an example of an approach that combines several of these processes within one computational framework. It connects models of electrical activity, muscle contraction, cardiac deformation, and blood flow. This allows the relationship between processes at the cellular level and the resulting behavior of the heart to be investigated. However, adding more physiological detail also increases the computational effort and introduces more parameters that must be estimated or validated using experimental or clinical data [SWH$^+$12].

Earlier computational studies frequently focused on individual parts of the heart, particularly the ventricles [BRF$^+$14]. More recent research has moved toward models that represent all four chambers and their interaction during the cardiac cycle [FPR$^+$23]. One example is the *Living Heart Project* [BRF$^+$14], which introduced an image-based model containing both atria and both ventricles. The model combines the electrical activation of the cardiac tissue with the resulting mechanical contraction of the heart. It can reproduce quantities such as cardiac deformation and pressure-volume relationships, which can be compared with clinical measurements. The project demonstrates how a whole-heart simulation can be used to study the combined behavior of the different chambers rather than examining each chamber separately.

Fedele et al. [FPR$^+$23] presented a more recent whole-heart model that includes additional physiological detail. Their approach considers the active contraction of both the atria and the ventricles, the orientation of the muscle fibres, differences between cardiac regions, and the connection between the heart and the circulatory system. The model reproduces several important characteristics of a healthy cardiac cycle, including changes in chamber pressure and volume, blood flow between the chambers, and the three-dimensional movement of the heart. The study also shows that atrial contraction can have an important influence on ventricular filling and should not always be neglected. Together the *Living Heart Project* [BRF$^+$14], *University of Tokyo heart simulator* [SWH$^+$12] and work by Fedele et al. [FPR$^+$23] demonstrate a part of the continuing development of computational models toward more complete representations of cardiac function. But despite their increasing level of detail, the two whole-heart models discussed represent the cardiac valves in simplified forms. In the *Living Heart Project* [BRF$^+$14] the valves are mainly represented through their effect on the flow of blood between the chambers rather than as individual deformable leaflet structures. Fedele et al. [FPR$^+$23] include the valves as simplified geometric regions within the openings between the cardiac chambers and vessels. These simplifications are appropriate because both projects focus mainly on the overall electrical and mechanical behavior of the heart. However, the models do not reproduce details such as the patient-specific shape, local deformation, or spatially varying thickness of individual valve leaflets.

Computational models are useful for studying the interaction between electrical activation, cardiac contraction, and blood flow. They can also be used to compare different physiological conditions and investigate processes that are difficult to observe directly. Nevertheless, their results depend on the assumptions, input data, and parameter values used to construct the model. In addition, a computational model cannot replace every type of physical experiment. Physical cardiac simulators and phantoms provide a complementary approach by creating a tangible model that can be used for experimental measurements and medical-device testing [Lee26]. This makes physical phantoms particularly relevant when the geometry and physical properties of a specific anatomical structure, such as the mitral valve of a single patient, is the main subject of investigation.

2.2 Comparable Pipeline Approaches

While the strengths of computational *in silico* models lie in their ability to simulate cardiac behavior reliably and reproducibly and to support clinical observations, the physical approach is motivated by several particularly interesting factors. Applications range from producing realistic replicas for teaching and training, to demonstrating operative procedures and treatments to patients and involving them in these processes, to presurgical planning and medical device testing, and even to printing implants for use in patient surgery. The latter has gained considerable traction in recent years and could offer substantial benefits if refined and widely adopted [TvO19]. The major distinction from computational models it that many measurements and observations can be made as well — although the behavior can be reproduced only experimentally — plus the additional benefits of possessing a tangible model which can be used for several other purposes as mentioned above [Lee26].

2.2.1 Source Data Acquisition and Preprocessing

In a 2019 review [TvO19], a PubMed search was conducted to assess the state of the art in 3D-printed cardiac valve models. The authors searched for English-language publications indexed up to January 2018 using the terms "*3D printing*" AND "*cardiac valve*", yielding 64 results. After screening for relevance and adding papers from reference lists, 29 publications remained [TvO19]. The same search in September 2026 yielded 262 results. This is an unfiltered count however, and the number of relevant publications is likely lower. Between February 2018 and September 2026, 198 items were added to the search results, suggesting a substantial increase in publications matching these terms.

Many of these studies required medical imaging data for model reconstruction, making it important to consider how these data are acquired and processed.

The creation of a patient-specific physical phantom begins with the acquisition of medical image data. The choice of imaging modality determines which anatomical structures can be visualized and modelled, as well as the additional processing steps required to produce a manufacturable model. A systematic review published in 2026 [PCCSLSN26] analyzed 63 studies on three-dimensional printing in mitral valve disease. Computed Tomography (*CT*) was the most commonly reported modality, used in 22 studies. TransEsophageal Echocardiography (*TEE*), a form of ultrasound imaging and the modality used for the data of this thesis, was reported in 16 studies. CT and TEE were used together in 10 studies, while Magnetic Resonance Imaging (*MRI*) was reported in only three.

Computed Tomography

The suitability of an imaging dataset for model production depends on sufficient spatial resolution, clear tissue contrast, and minimal artifacts and noise. Motion during CT acquisition can affect image quality, but cardiac and respiratory motion can be addressed through *ElectroCardioGram (ECG) gating*, *breath-hold* techniques, and/or *respiratory gating* in combination with a high-resolution scan [TvO19]. ECG and respiratory gating synchronize image acquisition with the cardiac and respiratory cycles, respectively, helping to reduce motion artifacts. Breath-hold techniques require the patient to hold their breath during the scan and can also reduce respiratory motion artifacts [GH13].

In general, CT imaging fulfils the requirements for producing a physical cardiovascular phantom, as it provides high spatial resolution and sufficient contrast for the segmentation of many cardiac structures. ECG-gated multiphase acquisitions can additionally be used to reconstruct the anatomy at a selected phase of the cardiac cycle [TvO19]. CT is particularly suitable for obtaining accurate measurements of cardiac structures and for visualizing calcifications. However, it involves exposure to ionizing radiation, may require the administration of a contrast agent [ZK21] and provides limited information about the tissue composition of the valve and myocardium [TvO19].

Ultrasound

Echocardiography is commonly used as the first-line imaging modality for the assessment of heart-valve disease. It is widely available and compared to the others it provides real-time information about valve morphology, leaflet motion, blood flow, and haemodynamic function. TransThoracic Echocardiography (*TTE*) is non-invasive and comparatively simple to perform, whereas TransEsophageal Echocardiography (*TEE*) can provide more detailed images of the valve and surrounding structures. In particular, three-dimensional transesophageal echocardiography can capture the spatial geometry and motion of the valve apparatus. However, TEE is a semi-invasive procedure that commonly requires sedation and may be unsuitable for patients with certain esophagus conditions [ZK21].

For the production of a patient-specific model, a three-dimensional ultrasound acquisition is required to obtain a volumetric dataset from which the anatomical structures can be segmented. The resulting image quality can nevertheless depend on the available acoustic window and the acquisition settings. Acoustic shadowing, image artifacts, and limited visibility of thin structures may make the boundaries of the valve leaflets difficult to identify [ZK21]. Consequently, the quality of an ultrasound-derived phantom depends strongly on the quality of the acquired volume and the reliability of the subsequent segmentation [TvO19].

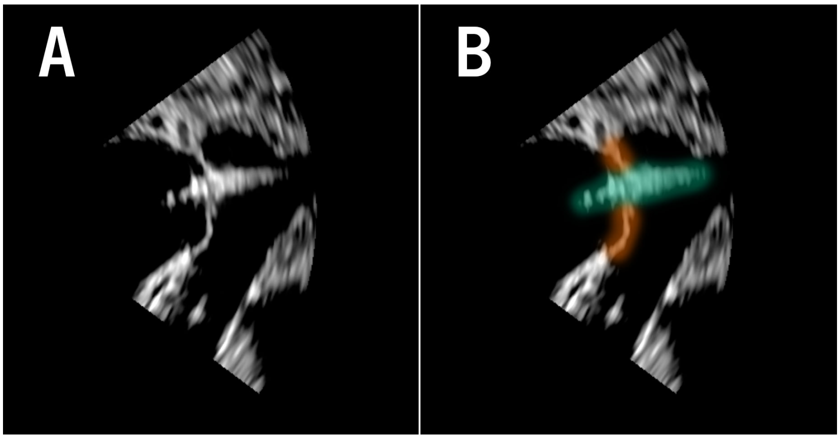


Figure 2.1: (A) Original TEE ultrasound image. (B) Color-coded image with visible sections of the mitral valve highlighted in orange and probe-related ultrasound artifacts highlighted in green.

To provide an example Figure 2.1 shows a coronal slice of the ultrasound data used to derive the model presented in this thesis, specifically the thickness measurements of the mitral valve leaflets, which is covered in Section 3.5. The figure also illustrates the challenge of reliably identifying and segmenting the leaflets in the presence of ultrasound artifacts and noise. *Image A* (left) shows the original, unaltered ultrasound image, while *Image B* (right) has been color-coded to highlight the visible portions of the mitral valve leaflets in orange and ultrasound artifacts associated with the TEE probe in green. The colored overlays are provided for visualization only and do not represent a segmentation.

TTE This is as mentioned the least invasive (as it is non-invasive) as the ultrasound probe is placed directly on the chest wall with a coating of gel in between to act as an interface for the ultrasound waves. No prerequisite preparation is required for the patient, it is highly available and it can be performed in a standard clinical setting at any time [May26]. TTE is limited by the available acoustic window and the presence of lung tissue as well as the ribcage that obstruct the ultrasound beam [PPL$^+$11]. Non-invasive ultrasound imaging is viable for quick and uncomplicated first-line assessments and monitoring behavior over time [ZAB$^+$17]. For our purposes however this modality remains fairly unsuitable as it is challenging to obtain a complete volumetric dataset with an emphasis for detail of the voxels in the region in and around the mitral valve [LBT+12].

TEE The modality for our ultrasound data and also an established modality for detailed assessment of mitral-valve anatomy is TEE [LBT+12]. The probe is inserted into the esophagus and positioned behind the heart, which allows for a closer view of the mitral valve and other cardiac structures [May26]. This proximity reduces interference from the lungs and ribcage, resulting in higher-quality images with better resolution [PPL+11]. However, TEE is semi-invasive and usually requires patient preparation, sedation, and monitoring during the procedure. Although the procedure is generally considered safe, potential risks should be assessed individually, taking the patient's overall health, age and underlying cardiac conditions into account [HAA+13]. TEE also entails greater equipment and logistical demands than TTE, including higher purchasing and maintenance costs and additional requirements for probe disinfection. Sedated examinations also require personnel for sedation and patient monitoring, together with postprocedural recovery [OGTB13].

ICE A type of echocardiography called IntraCardiac Echocardiography (*ICE*) has gained attention in recent years as it becomes more widely available and is increasingly used in clinical practice [ECD+25]. When performing ICE a ultrasound-tipped catheter is inserted directly into the heart introduced through usually the femoral vein [ECD+25]. This allows for high-resolution imaging of the heart and its structures from within the cardiac chambers [ESR+18], providing detailed visualization of the mitral valve and other intracardiac structures [ECD+25]. ICE is often used during interventional procedures, such as transcatheter valve repair or replacement, to guide device placement and assess procedural outcomes rather than for diagnostic purposes [ECD+25]. However, ICE is invasive and requires specialized training and equipment, limiting its use to specific clinical scenarios [ECD+25]. The cost of single-use ICE catheters can limit adoption [VMP+23]. Although ICE catheters are generally labelled for single use, they can be reprocessed and reused where permitted and appropriately regulated [PTS+26]. Reprocessed catheters may require replacement when image quality becomes inadequate or visible damage occurs [VMP+23]. Reprocessing can reduce catheter costs and facilitate broader use of ICE [PTS+26]. As this improved over the past years it is being adopted more widely. The procedure carries risks associated with catheterization, such as bleeding, infection, or damage to blood vessels [Bos18] although the need for ICE in certain scenarios can be justified [ECD+25].

Magnetic Resonance Imaging

Magnetic Resonance Imaging (*MRI*), also referred to as Cardiac Magnetic Resonance Imaging (*CMR*) when the cardiovascular system is scanned, is a non-invasive imaging modality that uses strong magnetic fields and radio waves to generate detailed scans of the body's internal structures [May23]. It is particularly useful for assessing structures of different soft-tissue types which would have lower soft-tissue contrast in CT imaging, allowing for an easier segmentation. Patients are, unlike CT, not exposed to harmful ionizing radiation. While there are no known risks associated with temporary exposure to the MR environment the strong magnetic field and radio waves carry specific risks [U.S].

The magnetic field will attract any ferromagnetic objects close to the scanner, which can cause injury to the patient or damage to the scanner. Patients with certain implants, such as artificial joints, stents, pacemakers or cochlear implants, may not be eligible for MRI scans due to the risks of interference with the device. The magnetic field will pull on ferromagnetic components and cause unwanted movement within the body. The combination of the changing magnetic field and the radio waves may cause heating of the implanted materials and can even cause burns in the surrounding tissue. Electronic devices may also malfunction or be damaged by the MRI and they can also degrade the image quality of the scan. Unless implants have been identified as MRI-safe or MRI-conditional these patients can not receive an MRI scan [U.S]. The procedure also can take up to an hour or more and requires the patient to remain still [May23]. Cardiac synchronization and breath-hold or respiratory compensation techniques can help reduce motion artifacts in the thorax and surrounding area of the heart during acquisition [FGMF13].

For the fabrication of a patient-specific mitral-valve phantom, however, the visibility of fine anatomical structures is particularly important. Although dedicated cine acquisitions can depict leaflet motion and identify prolapse or restriction [GSM17], conventional CMR provides less spatial and temporal detail than TEE which remains the preferred method for such assessments [FDMG+25]. Slice thickness and imaging-plane orientation also affect the depiction of thin leaflets. Consequently, visualizing a leaflet does not necessarily allow its local thickness to be measured reliably [GSM17].

2.2.2 Thickness Measurement

Going from the acquired medical image data to constructing a mesh model suitable for manufacturing requires knowledge of how the thickness of the mitral valve is distributed across the leaflets [VMGAL22]. The thickness of the leaflets is an important factor in determining their mechanical properties and haemodynamic behavior, as well as their interaction with the surrounding structures [LMM+23]. Unlike measurements of the overall length of the leaflets or annular dimensions, the thickness concerns a relatively small distance between very two close spaced boundaries. A reliable estimate therefore depends on whether these boundaries can be correctly identified and uniformly distinguished in the acquired image and crucially how missing or ambiguous boundary information in unclear regions are treated [XMC+18]. Previous studies span a range of different approaches, from manual measurements [HLM+20], overall averaging [ETZA+21] as well as a more automated pipeline [LVHV+23].

Hunter et al. [HLM+20] investigated the reliability of mitral valve thickness measurements.

The study does not provide insight to any manufacturing processes but solely the extraction of thickness measurements from two-dimensional echocardiography and whether they can be reliably repeated. The authors compared reader-optimized measurements, in which the reader selected an appropriate frame from a cine loop, with specialist-optimized images selected by an experienced specialists — as for this work only a single timeframe was provided this could not be taken under consideration. They also evaluated whether image magnification improved measurement reliability following established echocardiographic guidelines, by identifying and measuring the thickest portion of the anterior mitral leaflet during diastole. The study found that inter-reader reliability was not particularly high which demonstrates the difficulty of consistently identifying the leaflet boundaries in echocardiographic images. The relevance here lies in demonstrating the uncertainty associated with echocardiography-derived thickness measurements and that manual measurements require patient-bespoke identification of measured locations.

El-Tallawi et al. [ETZA$^+$21] developed a method for quantifying mitral valve leaflet deformation and strain using three-dimensional TEE. The study investigated normal mitral valves and valves affected by mitral valve prolapse, with different degrees of mitral regurgitation. Patient-specific valve geometries were reconstructed by segmenting the annulus and leaflets, and leaflet deformation was subsequently quantified by tracking the movement of points on the reconstructed surfaces between mid-systole and end-systole. The leaflet contour was traced to determine its cross-sectional area, while the distance between the leaflet tip and annular insertion was measured to obtain its length. Mean thickness was then estimated by dividing the cross-sectional area by the leaflet length. The maximum thickness estimate obtained across the two views was selected for each leaflet, and the overall mean mitral valve thickness was calculated from the anterior and posterior leaflet measurements. This method provides a practical estimate of average leaflet thickness, although it does not directly quantify local thickness variations across the entire three-dimensional leaflet. Importantly, thickness was not incorporated as an explicit spatially varying field in the strain model. Instead, the reconstructed leaflet surfaces were used to calculate strain from their deformation over time. The principal advantage of this approach is that it combines patient-specific three-dimensional geometry with quantitative measurements of leaflet deformation. However, the thickness estimate depends on manual contour delineation and image quality, and the surface-based strain analysis does not directly provide through-thickness strain or tissue stress. Consequently, the method is useful for estimating representative leaflet thickness and studying dynamic deformation, but additional geometric and material modelling is required to develop a fully volumetric mechanical model alongside a particularly labor-intensive pipeline.

Lopes et al. [LVHV$^+$23] proposed a particle-system approach for segmenting mitral valve leaflets from three-dimensional TEE. Their method identifies image features corresponding to the leaflet surfaces and investigates whether particle strength can provide information about the spatial distribution of leaflet thickness. In simple terms particle strength is a value describing the strength of an image feature detected by an algorithm, which Lopes et al. used as a clue to estimate the local thickness of the mitral valve leaflet. To generate a thickness representation, the authors assigned a linear relationship between particle strength and local leaflet thickness, using a maximum thickness of 5 mm based on published literature. They compared the resulting thickness estimates with manually segmented leaflet volumes and reported a root mean square (RMS) error of 0.99 ± 0.27 mm, meaning this method is assumed to be accurate. The authors noted that the relationship between particle strength and thickness requires further investigation, particularly because image contrast, chordae tendineae, and segmentation inaccuracies can affect the estimates. Nevertheless they mention that particle systems could become highly relevant for future work in this context as the manual labor can be cut short compared to uniquely picked measuring anchors.

2.2.3 Physical Model Fabrication

How the data is turned into a complete model heavily depends on the means utilized to retrieve that data. The step can range from a very trivial operation as in simply mapping values onto an existing mesh to having to create a new mesh from scratch [XMC+18, MPAYMJ+12, LVHV+23], which in both cases the process in doing so is dependant of and unique to the chosen respective software. As the final print is of interest, the different approaches of applying a thickness to the mesh [Ble26, LMM+23] are not discussed in detail here as there are too many different ways to do so and do not yield any effects on the output. The focus of this thesis lies on creating a pipeline while asserting a mesh is already provided. As the mode of the actual manufacturing method of a prototype however can be narrowed down to two main approaches — mold-based fabrication and additive manufacturing [SNI+18, IBN+22] — and each respectively the choice of a limited set of materials [MCZS19], I want to elaborate further on these aspects.

Mold-Based Fabrication

Molds allow the fabrication with the injection of a silicone. Although silicones and flexible polymers suited for 3D-printing exist, these are more costly and also require a high quality or even specialized printer to fabricate items in an acceptable quality [SNI+18]. Molds therefore offer a low-cost alternative in situations where perhaps the technology is not available or cost is a limiting factor. The negative mold can be constructed easily using a rigid polymer which in the case of most materials can be done from common desktop printers. PolyVinyl Alcohol (*PVA*) which dissolve in water behaves similar during printing as most rigid polymers. Using this to print the negative and then cast the valve in silicone allows the surrounding mold-structure to be diluted away resulting in the finished valve remaining. This makes each mold single-use but the combined material costs lie at roughly US$2 and the time to produce one complete model at around 8,5h, 6h of which for the PVA to dissolve entirely, making the process of fabricating a valve straight-forward and accessible [PMF+18].

Additive Manufacturing

The most common techniques for additive manufacturing of mitral valve disease can be categorized into three main groups: Fused Deposition Modeling (*FDM*), StereoLithography (*SLA*), and PolyJet printing [PCCSLSN26]. FDM is a widely used technique that involves the extrusion of thermoplastic filaments layer by layer to build a three-dimensional object. It is relatively inexpensive and accessible, but it may have limitations in terms of resolution and surface finish [MCZS19]. SLA uses ultraviolet light to cure liquid resin layer by layer, resulting in high-resolution prints with smooth surfaces. However, SLA printers can be more expensive and require post-processing steps such as washing and curing [HQW20]. PolyJet printing is a high-end technique that jets layers of liquid photopolymer onto a build tray and cures them with ultraviolet light. It allows for multi-material printing and can produce highly detailed models with varying mechanical properties, but it is also more costly and requires specialized equipment [PCSS22].

A case was documented at the Stanford School of Medicine of a patient who already had a mechanical mitral valve replacement and was suffering from a leak between the left atria and left ventricle. The patient had to undergo a procedure to close the leak. The team used a 3D-printed PolyJet model of the patient's heart to plan the procedure and select the appropriate method of closure. The model allowed the team to visualize the anatomy of the leak which helped the radiologists and surgeons to choose the best approach and avoid complications. Despite this case emphasizing the modelling of the heart and structures around the valve and not the valve itself, this nevertheless demonstrates that 3D-printing the exact geometry of a patient to assess specific clinical situations can help in the planning of surgery [Sta].

All of the three methods mentioned have the capability to produce tangle models made up of a flexible material. FDM printing is generally used to create functional prototypes as it lacks the ability to fabricate intricate details compared to SLA or PolyJet. However it is utilized in the production of tools and support structures used in surgery and it is sufficient enough to make (not fully functional) models of the mitral valve [PCCSLSN26]. In our case this is of special interest as it lies within the scope of this project to analyze to which extent a FDM-printed model can simulate and/or imitate the cardiovascular system in the context of ultrasound imaging.

2.3 Identified Gap in Literature

A search of the current literature (November 2026) reveals that while there are several studies on computational cardiovascular simulations and physical phantoms as covered within this chapter, there is a lack of research specifically focused on a FDM-printed mitral valve phantom which aims to behave similar to the real mitral valve in ultrasound imaging. Most of the research found based on the goal of producing a physical model aims to replicate the geometry of the mitral valve and surrounding structures for surgical planning and device testing. However, there is limited research on the development of a patient-specific mitral valve phantom that can be used for ultrasound imaging and training purposes as well of closely resembling a patients anatomy. This gap in the literature highlights the need for further research on the development of a FDM-printed mitral valve phantom that can accurately replicate the geometry and mechanical properties of the real mitral valve in ultrasound imaging. FDM-printing is hereby the chosen method of fabrication as it is a low-cost and accessible technique that can be used to produce patient-specific models.

CHAPTER 3

Methodology

3.1 Overview of the Approach

The aim of this work is a 3D-printed prototype of a thickness-accurate 3D model of the valve leaflets that can serve as the basis for an ultrasound phantom. We start with a volumetric ultrasound scan of the valve, stored as a VDB volume and a surface mesh of the leaflets. Unfortunately these two do not provide all parameters that are necessary for the production for a full tangible model. The scan contains the full valve, but only as a grid of grayscale values with no explicit geometry. The mesh has clean geometry, but it is only a single surface with no thickness, so on its own it cannot be converted into a physical object with a wall.

The core task of the whole pipeline is therefore to bridge these two: to bring the mesh and the volume into a common coordinate system, to recover the leaflet thickness from the volume, to transfer that thickness to the mesh, and to turn the single surface into a closed solid.

I did not utilize a single program to perform these steps, as no single tool performs well at every part. Instead the work is split across three platforms and six major steps — also denoted in Figure 3.1: MeVisLab [MeV] and Blender [Ble] are used to load and prepare the volume and mesh (Step 1 described in Section 3.2), 3D Slicer [FBKC$^+$12] is used to align the mesh with the volume (Step 2 Section 3.4) and to take the thickness measurements (Step 3 Section 3.5), and Blender is used to interpolate the measurements over the mesh (Step 4 Section 3.6) and build the solid (Step 5 Section 3.7). This solid is then printed to provide a physical model (Step 6 Section 3.8). Minor procedures, mainly the interpolation, I have implemented with short Python scripts using the numpy library. The following sections describe each part and why each choice was made and not one of the other possible ones.

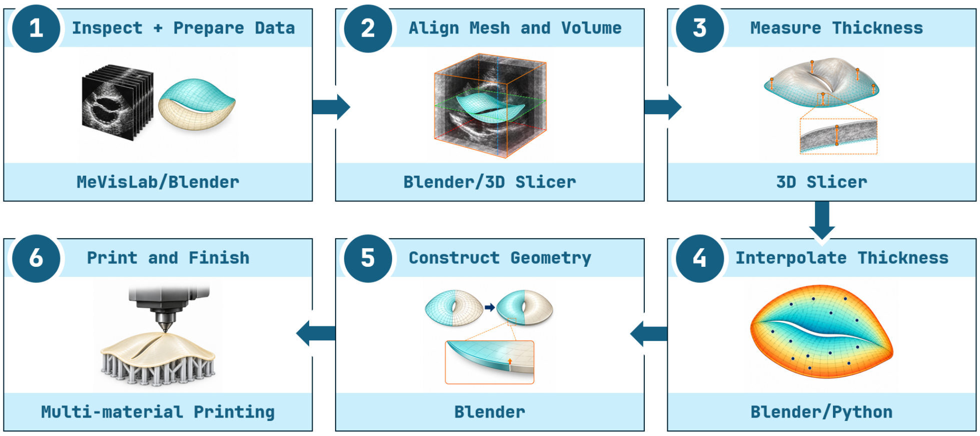


Figure 3.1: Mitral valve reconstruction and fabrication pipeline schematic. Illustrations generated using ChatGPT from methodological descriptions and refined through iterative prompting [Ope26].

3.2 Input Data

The first input is the volumetric scan. It is a 3D grayscale field where bright voxels roughly correspond to tissue and dark voxels to the surrounding space. It contains the leaflets with their real thickness, but this thickness is only present as a bright band in the image and not as a number or a shape that can be measured automatically without further steps. The scan also contains impurities at several areas: the ultrasound probe used to record it left a very noisy region in the centre of the valve, around the opening. How we can deal with this will be discussed throughout this chapter.

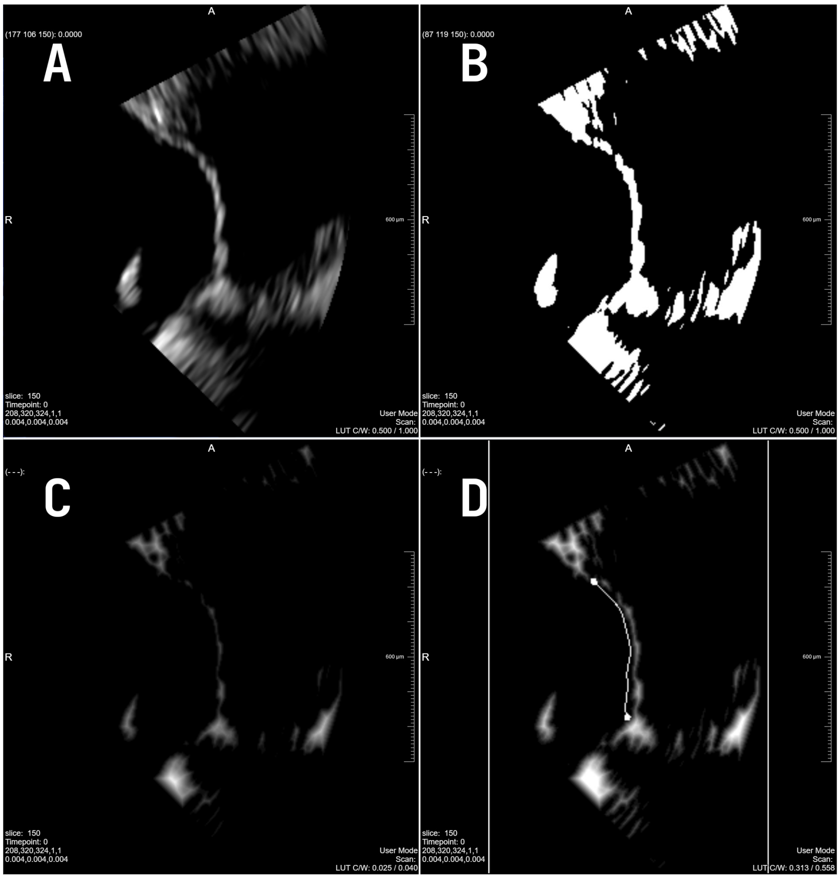
The second input is the surface mesh of the leaflets. This mesh is a single surface. It does not enclose a volume, and it has no wall thickness at all. In practice there are two separate leaflet meshes, referred to here as [Mesh1] and [Mesh2].

Before deciding how to add thickness, I first had to establish what the surface actually represents, because this defines how the thickness is calculated and applied to the mesh. A surface mesh of a thin structure can either be a *midsurface*, sitting in the middle between the two walls, or a *boundary surface*, sitting on one side. If it is a midsurface, the real thickness is split evenly on both sides of it. If it is a boundary surface, the whole thickness sits on one side only. Misinterpreting this would make all thickness values off by a factor of two.

To validate this, the mesh was placed next to the volume and a cross-section through a leaflet was examined. If the mesh were a midsurface, it would run through the middle of the bright leaflet band. In this case the surface sits on one side of the band, which means it is the underside (a boundary surface), *not* the middle — depicted in Figure 3.2 (D) as well. This has a direct consequence for the later steps: the full measured thickness has to be added on one side, and there is no division by two.

3.3 Modality of Thickness Recovery

Thickness is not stored anywhere in the data. It has to be recovered from the grayscale volume. There were two reasonable ways to do this, and this sections explains both because the reason for choosing the second one is part of the method.


A
(177 106 150): 0.0000
A
R
600 µm
slice: 150
Timepoint: 0
208,320,324,1,1
0.004,0.004,0.004
User Mode
Scan:
LUT C/W: 0.500 / 1.000
A
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600 µm
slice: 150
Timepoint: 0
208,320,324,1,1
0.004,0.004,0.004
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0.004,0.004,0.004
User Mode
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LUT C/W: 0.025 / 0.040
A
(- - -):
D
R
600 µm
slice: 150
Timepoint: 0
208,320,324,1,1
0.004,0.004,0.004
User Mode
Scan:
LUT C/W: 0.313 / 0.558

Figure 3.2: (A) Original ultrasound image. (B) Ultrasound image with a threshold set at >0.2. (C) Ultrasound image after the Euclidean Distance Transform. (D) Mesh mapped onto the Euclidean Distance Transformed image — all images captured from an identical slice

The initial concept for the pipeline was to compute the thickness fully automatically with a distance transform. The steps would be as depicted in Figure 3.2: turn the unaltered volume (A) into a binary mask (B), where material is 1 and everything else is 0, and then run a distance transform on that mask. A distance transform gives, for every inside voxel, the distance to the nearest background voxel, in other words the distance to the closest wall resulting in image (C) [FPWW]. In the middle of a leaflet, that distance is half the local thickness, so the thickness could in principle be read off as two times the distance value. This was set up in MeVisLab using an `OpenVDBLoad` module to load the volume, a `Threshold` module to make the binary mask (values above about 0.2 set to 1, everything else to 0), and a `EuclideanDistanceTransform` module in 3D mode to produce the distance field. Visible in image (D) the mesh is not a midsurface as it is positioned on the underside.

The reason this idea is promising is that it is automatic and dense: it would give a thickness value everywhere on the leaflet at once, without any manual work. However I decided to abandon this method, considering the following points:

First, there is a practical problem with reading the result. To get thickness at the mesh, the distance field has to be sampled exactly at the positions of the mesh points. MeVisLab does not provide a ready-made module for "sampling from volume X at the vertices of mesh Y", and the scripting module has no image input or output that would let this be done directly inside the network. It would mean exporting the distance field and doing the sampling externally in Python, which is possible however adds another step that would itself have to be checked.

Second, the simple "two times the distance" rule only holds if the mesh is a midsurface. As shown in Section 3.2 and depicted in Figure 3.2 (D), the mesh here is the underside. For a boundary surface the correct value would be the distance from the underside across to the top wall, which is not what a symmetric distance transform gives without extra work. So the clean version of the method does not actually match this data.

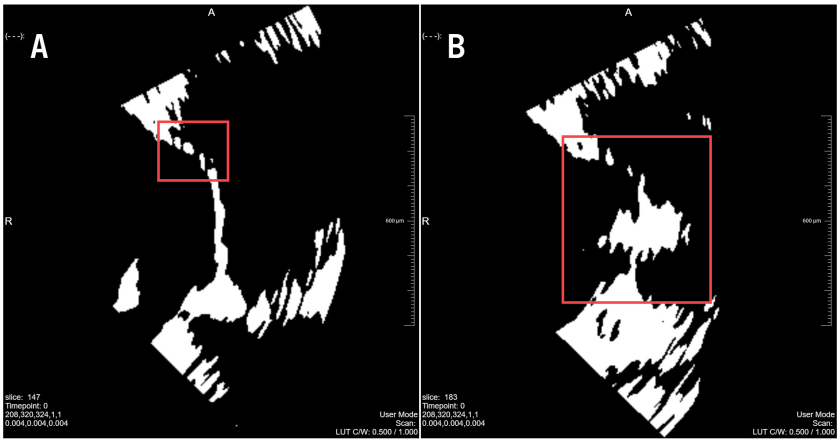


Figure 3.3: Thresholded ultrasound slices illustrating segmentation artifacts in the valvular material. (A) Discontinuity of the segmented membrane near the valve edge. (B) Substantial noise and false segmentation within the central region. Regions of interest are highlighted by red boxes.

Third, the whole result depends on one global threshold that separates material from background and given the noisy volume this separation is not trivial. Figure 3.3 helps to illustrates the issue. Image (A) shows the interrupted tissue of the valve. The Ultrasound volume has speckled and blurred edges, and this scan has an especially noisy region in the centre of the valve as can be seen in image (B). The highlighted region marks the area where the valve is meant to be but due to severe noise it cannot be identified. A single fixed threshold applied to the whole volume turns that noise into false geometry, especially in noisy regions, and there would be no simple way to reliably tell noise from tissue after thresholding. Changing the threshold does not solve the issue since lowering it would result in perhaps more continuous regions but also artifacts not actually within the leaflets therefore resulting in larger thickness readings than the true value. Increasing the threshold results in even less surface area to measure from. Resolving these difficulties with preprocessing and interpolation would be possible to some extent but again add several steps to the entire pipeline.

Taken together, these points describe a method that would contain a lot of parameters and numbers that could not be reliably validated. For the extent of this thesis, a dense automatic result that cannot be verified is less desirable than a smaller set of values that can. This is the reason I chose to measure the thickness manually at a limited number of well-imaged locations instead. This however, only works if the mesh is correctly placed inside the volume, so the first step before measuring was to align the two entities.

3.4 Aligning the Mesh and the Volume

The mesh and the volume did not overlap when they were first loaded together in both MeVisLab and 3D Slicer — only Blender could directly open the .usda file which preserved both coordinate systems in the correct orientation. In MeVisLab and 3D Slicer they sat in different coordinate systems. This alignment matters here because, as described in Section 3.5, the thickness is measured along lines perpendicular to the mesh surface. If the mesh is offset, those perpendicular lines start offset, and the measurement is not precise. So the mesh and the volume had to be brought into the same coordinate frame first. Before performing any geometric operations in Blender, all input data was uniformly scaled up by 750%. This initial upscaling was applied as for one to improve numerical precision during computational and transformation steps within Blender's modeling environment (as further discussed in Section 3.6) and resolved export scaling issues I encountered, ensuring the resulting STL file imported at a realistic scale in the 3D-printing slicing software. This is not proposed as a general scaling factor for other datasets as in an ideal scenario the mesh and volume are at the right scale. Taken under consideration that the scaling could have been off by a factor of 100 as perhaps there was a miscommunication in metric units (centimeters in meters) between the slicing-software and the modeling environment upscaling by 750% seamed very reasonable and was ultimately chosen as this resolved in a similar scale as the reference values determined by Dwivedi et al. [DMJ$^+$14]. Furthermore this uniform scaling is fully tolerable for the valve model, as its final absolute dimensions are ultimately constrained by the physical dimensions of the ventricular chambers developed by Machoritsch, allowing the valve's scale to be quickly adjusted accordingly prior to final production.

Aligning the two by their shapes directly is difficult, because the leaflet surface is smooth and curved and offers no clear matching landmarks to either decide on an aligning transformation nor check if they are overlapping correctly. The latter being particularly crucial. Instead, I used a reference that all three programs can interpret correctly: the bounding box. MeVisLab, 3D Slicer and Blender each draw an axis-aligned bounding box around the volume, and this box is the same in all of them because it is derived from the same volume. Since everything was perfectly aligned in Blender I added planes to the mesh that correspond to the bounding box of the volume, so that the mesh carries the same box as a built-in reference. The resulting mesh is depicted in Figure 3.4. Matching the mesh's bounding planes to the volume's bounding box then defines the alignment completely, because the two boxes have to coincide.

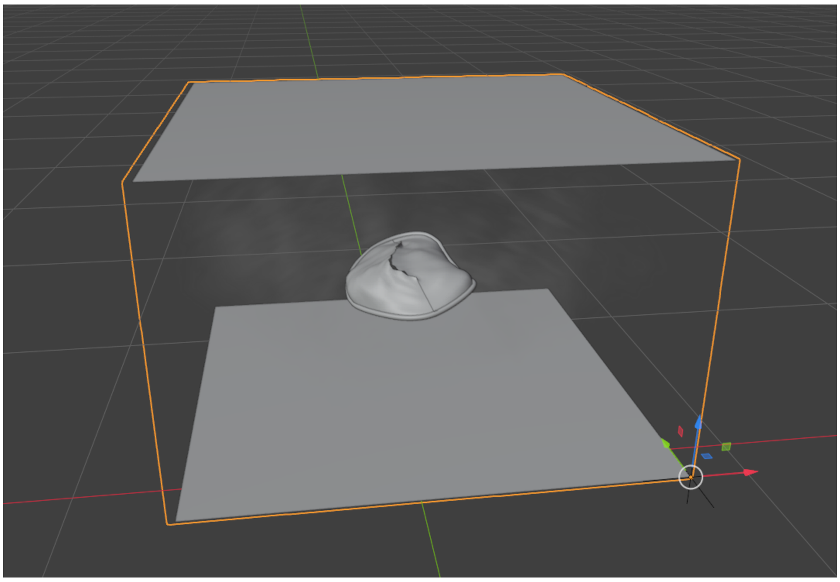


Figure 3.4: Planes added to the mesh in Blender to correspond to the bounding box of the volume.

From this box-to-box correspondence a transformation matrix was obtained, which was applied to the mesh in 3D Slicer. After applying it, the mesh is aligned inside the volume in the correct position and orientation, so that a leaflet in the mesh corresponds to a bright band in the scan. Using the bounding box as a shared reference is more reliable here than utilizing a method which registers the curved surfaces against each other, and it keeps the alignment reproducible, because the box is defined by the data and not by a subjective judgement of where the surfaces should meet.

3.5 Measuring Leaflet Thickness in 3D Slicer

With the mesh aligned, the thickness was measured in 3D Slicer [FBKC$^+$12]. The tool used is the line (ruler) markup, which gives a direct distance reading in millimeters between two points. The important part here is the rule I followed for placing the two points achieving repeatability for the measurements.

I did not to determine the leaflet edges purely by eye, and I also chose not to threshold the whole volume, for the reasons given in Section 3.3. Instead, I used a local intensity criterion, applied only at the spot being measured. As an acceptable tradeoff, considering the available data and its quality, I selected to interpret everything from an intensity of 0.2 and upward as actual tissue as this offered a decent amount of area to take measurements from but would also not overestimate on tissue which clearly was not part of the mitral valve. Each measurement line was placed perpendicular to the mesh surface at the chosen location. Along that line, the measurement starts where the intensity first rises to 0.2 and ends where it falls back on or below 0.2, and the thickness is the length of that span. This is the same kind of intensity criterion that the automatic approach would have used, but applying it locally, along a single line at a spot I had judged to be clean, avoids the problem of applying it to the whole noisy volume at once. I chose where to place each line; the intensity rule then decided where the tissue began and ended.

Where the measurements were taken was itself a choice driven by the data quality. Thickness is not the same everywhere on a leaflet, so I wanted to sample across the surface, but two regions in particular are challenging to measure. The first is the centre of the valve, around the opening, where the probe left a noisy hole in the scan. In that region it is difficult to see a clean tissue band, so reliable perpendicular measurements are not easy to obtain. The second is the thin outer brim of the leaflet, at the edge of the mesh, where measurements were also difficult to take cleanly. In between these two problem regions the data is clearer, so most of the measurements were taken there, where the tissue band is well defined and the intensity rule can be applied with confidence. The resulting locations where measurements were taken are shown in Figure 3.5 although the numbers in the figure do not correspond to the actual indexing of the measurements in Table 5.1. It should be mentioned that the aortic valve is shown as well in the figure and it was within the scope of the project to produce a model of the aortic valve too, but due to the poor quality of the ultrasound data around it there were nearly any areas where measurements could be taken. Therefore, the aortic valve was not included in the final model.

For the two difficult regions I made a deliberate simplification rather than forcing measurements the data does not support. For the noisy centre I sampled all the measurements I could obtain in this region and selected the median and treated the thickness across the whole centre region as constant at that value. I did the same for the outer edge, selecting the median of the edge measurements and treating the edge as having one constant thickness. The median was used rather than the mean because it is less affected by bad readings, which is exactly what is expected in these noisy regions. Assuming a homogeneous thickness in the centre and at the edge is a simplification, and it is stated as such, but it is a reasonable in this case: the data in those regions is not good enough to justify anything finer, and a single well-chosen value is more justifiable than a set of unreliable point measurements.

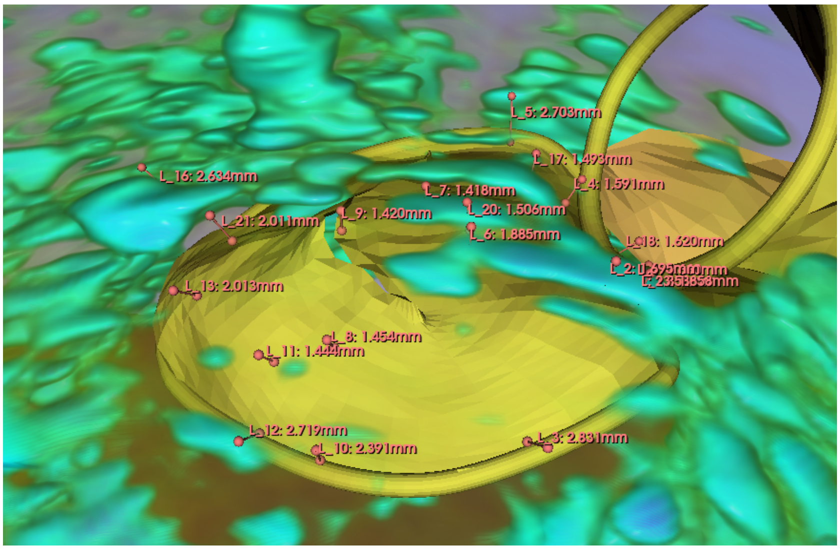


Figure 3.5: Valve mesh with volume and locations of the manual thickness measurements on the leaflets meshes from 3D-Slicer. Mesh depicted in yellow, ultrasound volume in blue/turquoise and measurement points in red

3.6 Spreading the Measurements over the Mesh

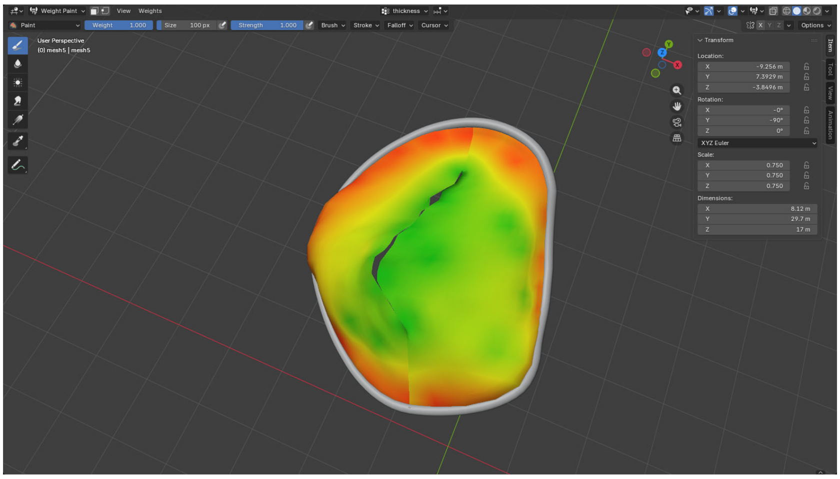
For each leaflet I obtained a set of thickness values: several point measurements in the well-imaged middle region, plus one constant value for the noisy centre and one constant value for the outer edge. The mesh, however, has several thousand vertices, and the step that builds the solid (Section 3.7) needs a thickness value at every single vertex. So the measured values have to be distributed over the whole surface.

The measured points can be recorded as follows in Table 5.1, which lists the coordinates of each measured point and the corresponding thickness value. The table is split into two parts, one for each leaflet mesh, and it includes the sample points, the outer rim points, and the centre points. The outer rim and centre points are treated as having constant thickness values, as described in Section 3.5.

The simplest way to solve this would be to give the whole leaflet one average thickness. I rejected this because it would remove the variation between the base and the edge, resulting not in a desired realistic model. Therefore a type of interpolation is needed for the region that is well sampled, while the centre and the edge keep their constant median values as described in Section 3.5.

For the interpolation I chose inverse distance weighting (*IDW*) [She68]. The idea is easy to state: each vertex is given a thickness that is a weighted average of the known values, and values that are closer to the vertex have more significance than ones further away. The weight of each value falls off with distance, following one over the distance raised to a power. A vertex located right next to a known value gets almost exactly that value, and a vertex halfway between two of them gets roughly their average.

I chose IDW over more advanced interpolation methods such as kriging or radial basis functions. With only a finite set of known values, the more advanced methods bring in extra assumptions and parameters that could not be checked with so little data. IDW, in contrast, makes no assumption about an underlying function, works with irregularly placed values, simple and perfectly adequate in this case. For the power parameter, a value of two was used. This gives greater weight to nearby measurements while still providing a smooth transition between the measured points. The resulting assignment of thickness values to the mesh is shown in Figure 3.6. The outer brim of the leaflet is shown in red as it has the maximum thickness, while the centre of the leaflet, the thinnest part, is shown here in green which is roughly half of the maximum thickness and therefore not in blue, as blue would map to a thickness of zero. The transition between the two is smooth, as expected from the interpolation.



Weight 1.000
Size 100 px
Strength 1.000

Transform

Location:
X -9.256 m
Y 7.3929 m
Z -3.8496 m

Rotation:
X -0°
Y -90°
Z 0°
XYZ Euler

Scale:
X 0.750
Y 0.750
Z 0.750

Dimensions:
X 8.12 m
Y 29.7 m
Z 17 m

Figure 3.6: Top-down view of the leaflet mesh in Blender, showing the interpolated thickness values as a vertex group. The color gradient represents the thickness, with red indicating thicker regions and blue indicating thinner regions.

The interpolation itself was carried out in Blender using a short Python script, as shown in Listing 5.1 and the values and coordinates as in Table 5.1. NumPy [HMVDW$^+$20] is used for the distance and weighted averaging calculations. For each vertex, the script calculates the Euclidean distance to all measured points, determines the corresponding inverse-distance weights, and computes the weighted average of the measured thickness values. For every vertex the script computes the distances to the known values, forms the weights, and takes the weighted average. The result is stored on the mesh as a per-vertex weight, called a vertex group in Blender. Before storing, the values are scaled to the range zero to one by dividing by the largest thickness, because the modifier used in the next step expects weights in that range.

One limitation of this choice should be stated openly. IDW here uses straight-line (euclidean) distance through space, not distance measured along the curved surface. Across a fold, or across the gap between the two leaflets, straight-line distance could mix values that should not be mixed, because two points can be close in space while being far apart along the tissue. Figure 3.7 helps to clarify this issue. When performing IDW with no distinction between leaflets the weight given point C would be influenced more by point A than by point B as point A is closer in space. This is exactly why the two leaflets are kept as two separate meshes and interpolated separately: a value measured on one leaflet [Mesh1] cannot leak across the gap onto the other [Mesh2]. Also within a single, fairly flat leaflet and given a reasonably close spread of measurements, the difference between straight-line distance and along-surface distance is small enough to accept for the purpose of this work. As the dashed line representing the distance from an interpolated vertex (D) to a given point (B) is fairly close to the actual shallow curvature of the mesh.

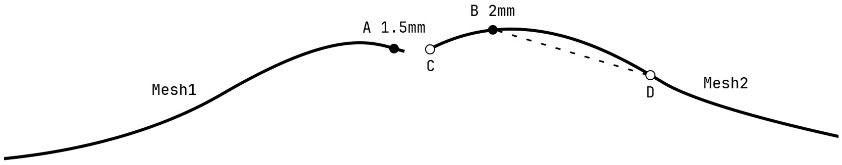


Figure 3.7: Schematic illustrating the use of Euclidean distance in IDW thickness interpolation. Although vertex C lies on [Mesh2], it is closer to measurement A on [Mesh1] than to measurement B on its own leaflet. Interpolating the leaflets separately prevents this influence across the gap. Within [Mesh2], the dashed straight-line distance between B and D closely approximates the distance along the shallow surface curvature.

3.7 Constructing the Solid Model

We now have a thickness value at every vertex, and a single surface that represents the underside of the leaflet, and this has to be turned into a closed solid whose wall has the correct local thickness.

For this step, Blender's Solidify modifier was used [Ble26]. This modifier takes a surface and extrudes it by a chosen thickness to produce a solid, and, importantly, it can scale that thickness per vertex using a vertex group. This is what makes it fit the situation here: the interpolated thickness field from Section 3.6 is stored as exactly such a vertex group, so it maps directly onto how far each part of the surface is pushed out. In practice the modifier's thickness is set to the largest thickness value, given in metres because Blender works in metres, and the zero-to-one weights then scale every other part of the surface below that maximum. The local thickness at a vertex is the weight at that vertex multiplied by the modifier thickness.

The extrusion is done in one direction only, which is where the check from Section 3.2 finally pays off. Because the mesh is the underside of the leaflet and not the midsurface, the whole thickness has to be added on one side, growing outward from the surface, rather than split half on each side. This is also the concrete reason why the "two times the distance" rule from the distance-field idea in Section 3.3 would have been wrong for this data: a factor of two is only correct when the surface is the centre, and here it is a boundary. In the modifier this is set by using a one-sided offset so that the material grows only outward. The direction of the extrusion follows the surface normals, so the normals were checked beforehand and flipped where needed, to make sure the wall grows away from the leaflet body and not into it.

Each leaflet is solidified on its own before the two are combined. The reason is that the two leaflets have different thickness ranges, and a single modifier can only hold one thickness value and one vertex group. So [Mesh1] and [Mesh2] are each given their own modifier with their own maximum thickness. Only after the thickness has been fixed into the geometry, by applying the modifier, are the two meshes joined into one object. Joining them earlier would force a single shared thickness on both, and would also let the interpolation bleed across the gap between them, for the reason given in Section 3.6. The output of this step is a single solid mesh in which the leaflet walls have the measured, locally varying thickness.

3.8 Printing the Complete Model

Once a sufficiently accurate approximation, or model, of the mitral valve had been successfully developed, the actual objective and core of this work was addressed: the fabrication of a realistic ultrasound phantom. Fused Deposition Modeling (*FDM*) 3D printing — works by layering an extruded filament over itself — was selected as the manufacturing method [MCZS19] as it allows for a fairly quick and uncomplicated production of a prototype. With an established pipeline for retrieving the model it can be scaled to different production platforms. Modern 3D printers provide sufficient detail resolution and, importantly for this project, the ability to work with flexible materials such as Thermoplastic PolyUrethane (*TPU*). The fabrication of flexible structures using molds and a material such as silicone would also have been possible and could even have enabled scalable production from the start. However, 3D printing was ultimately chosen because it allows for easy modifications to the model and eliminates the need to consider how the mold can be removed after casting.

Throughout the project, while the thickness measurements were being determined, test prints utilizing an *Original Prusa i3 MK3S+* were conducted in parallel to identify the approach that would yield the best possible result. It was apparent that the heart valve would require support structures during printing, as the convex geometry would need to rest as flat as possible on the build plate, resulting in overhangs of well above 60° [MCZS19]. Printing the model vertically slightly reduced the cross-sectional area requiring support, but not to a significant extent. Furthermore, this orientation caused the structure to become unstable during printing, as the build plate moves back and forth along the y-axis. The higher the print progressed, the more pronounced this movement became, causing the structure to wobble.

A major disadvantage of support material is the difficulty of completely removing all residues and achieving a smooth surface after printing. Test prints using PolyLactic Acid (*PLA*), for example, which is a relatively rigid polymer, demonstrated that removing the support structures alone was already a considerable challenge. Achieving a surface as smooth as the upper side of the print would then require substantial additional effort. Printing with flexible TPU did not significantly change this situation. The TPA95A [Reca] used was specified with a Shore A hardness of 95; in general, lower Shore A values correspond to softer and more flexible TPU materials — e.g. TPU95A is flexible but fairly stiff while TPU60A is more flexible. It could be printed using settings that were almost identical to those used for PLA. For TPU82A [Recc], the printing speed was reduced to avoid compromising print quality. However, at the desired thickness of the heart valve, this material remained too inflexible. TPU60A [Recb], which provided an appropriate degree of flexibility and softness, introduced another problem: the amount of support material used previously was insufficient to adequately support the material between the individual support structures. The resulting print contained numerous holes and areas where the printed material had fused with the supports. The solution was to increase the density of the support structures by reducing the spacing between them. In this case, this was achieved by selecting smaller squares in the grid pattern.

This resulted in a print using the desired material that appeared intact at first glance. However, the issue of the difficult-to-remove support structures remained. To address this problem, additional prototypes were produced using an *Original Prusa XL5T*, a multi-nozzle printer capable of printing with multiple materials simultaneously. The idea was to print the supports using a water-soluble filament, in this case PolyVinyl Alcohol (*PVA*) [MKM⁺23], while continuing to print the mitral valve using TPU60A. The finished model could then have been placed in water, allowing the supports to dissolve completely without leaving any residue.

However, there were concerns that the TPU might not adhere sufficiently to the PVA supports during printing, potentially resulting in print defects in the model. Therefore, PLA supports were initially used instead. Ultimately, this proved to be the right choice for this particular application, as the PLA and TPU adhered strongly enough to ensure a successful print while still allowing the valve to be easily peeled away from the support material after completion. As hoped, this resulted in a perfectly smooth underside. Furthermore, the gap within the valve opening was preserved. Although as for the print settings such as temperature, speed, retraction- and flow-rate, etc. all parameters were set as recommended by the filament manufacturer [Recb] working with a highly flexible filament resulted being tedious and required several attempts to produce a successful print. The TPU60A was prone to clog inside the extruder which resulted in failed prints as the material could not be applied fast enough as an example of this is displayed in Image 3.8. The final slicer settings are listed in Table.

make Table

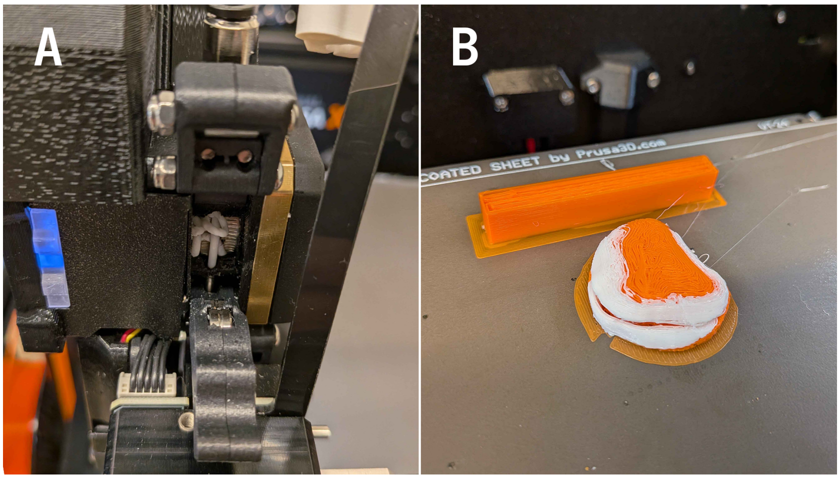


Figure 3.8: Pictures taken from 3D-printer's extruder (A) and print bed (B) displaying failed prints due to clogging of the extruder. The left image shows the extruder clogging in the gears, while the right image shows the print bed with a failed print. *Pictures taken by Dr. techn. Daniel Pahr*

The only remaining step was to manually remove any excess strands of filament and create a small incision in the opening using a scalpel. This was necessary because 3D printing small openings generally leaves behind thin strands of filament, a phenomenon known as *stringing*, which can cause the opening to become partially sealed [Pru]. The completed prototype, including the support structures, was handed over to Siemens Healthineers, where the valve underwent testing and evaluation by clinical specialists.

CHAPTER 4

Results

This chapter presents the results of the developed reconstruction and manufacturing pipeline. It briefly covers the resulting leaflet geometry, the fabrication of the physical prototype as well as its qualitative evaluation using ultrasound imaging by Siemens Healthineers and subsequent future use-cases for the findings.

4.1 Reconstructed Leaflet Geometry

The developed pipeline successfully transformed the initially unthickened meshes of the valve into a solid representation of the mitral valve leaflets with spatially varying thickness. The thickness values were obtained from manually selected locations within the volumetric ultrasound dataset and subsequently interpolated across the respective leaflet surfaces using inverse distance weighting.

A total of 46 locations are documented in Table5.1, with 23 locations assigned to each leaflet mesh. These include measurements from regions with clearly identifiable leaflet boundaries, as well as representative median-derived values for the central and outer rim regions, where the ultrasound image quality did not allow for equally reliable measurements. The resulting thickness values demonstrate a spatial variation across the leaflets. The manually sampled values range from 1.418 mm to 2.831 mm. For both leaflets, the outer rim was assigned a constant thickness of 2.634 mm, while the central region was assigned a constant thickness of 1.420 mm. The remaining values were interpolated across the respective surfaces.

The resulting distribution is illustrated in Figure3.6. As intended, the thickness gradually transitions between the thinner central region and the thicker outer regions. The two leaflets were interpolated independently to prevent measurements from one leaflet influencing the thickness distribution of the other.

The thickness fields were subsequently applied to the surface meshes using Blender's Solidify modifier. Since the original meshes represented the underside of the leaflets rather than their midsurfaces, the full thickness was extruded in one direction. This produced a solid geometry incorporating the reconstructed thickness distribution.

4.2 Physical Prototype

The resulting solid model was successfully manufactured using Fused Deposition Modeling (*FDM*) with flexible TPU60A filament.

The manufacturing process required several iterations to obtain a satisfactory result. Some attempts encountered problems related to filament clogging, insufficient support density, and difficulties removing support structures without damaging the printed leaflets. The final manufacturing approach used PLA support structures in combination with TPU60A for the valve itself. This allowed the support material to be removed while retaining the printed leaflet geometry.

The completed prototype retained the intended opening between the leaflets. Following the removal of support material and excess filament, the underside of the printed structure exhibited a smooth surface.

Figure 4.1 shows the completed physical prototype of the mitral valve from the top and bottom views. It can clearly be seen that the support structures were successfully removed, leaving the intended geometry intact. The leaflet surfaces appear smooth and continuous — by the standards of unaltered FDM prints.

The successful fabrication demonstrates that the reconstructed geometry could be transferred into a physical model using an accessible FDM printing process and a flexible thermoplastic material.

4.3 Qualitative Ultrasound Assessment by Siemens Healthineers

Following manufacturing, the physical prototype was provided to Siemens Healthineers for evaluation by clinical specialists.

The feedback was provided by a clinical expert stationed in Vienna. In their response, they confirmed that the newly printed valve was clearly visible using ultrasound imaging and described it as nicely scannable.

The valve was placed in a medium of water and scanned using the *ACUSON Sequoia Ultrasound System*. Photographs of the ultrasound equipment display were provided alongside this feedback. These photographs show the ultrasound images obtained during the examination of the physical prototype and serve as qualitative evidence of its visibility in ultrasound imaging.

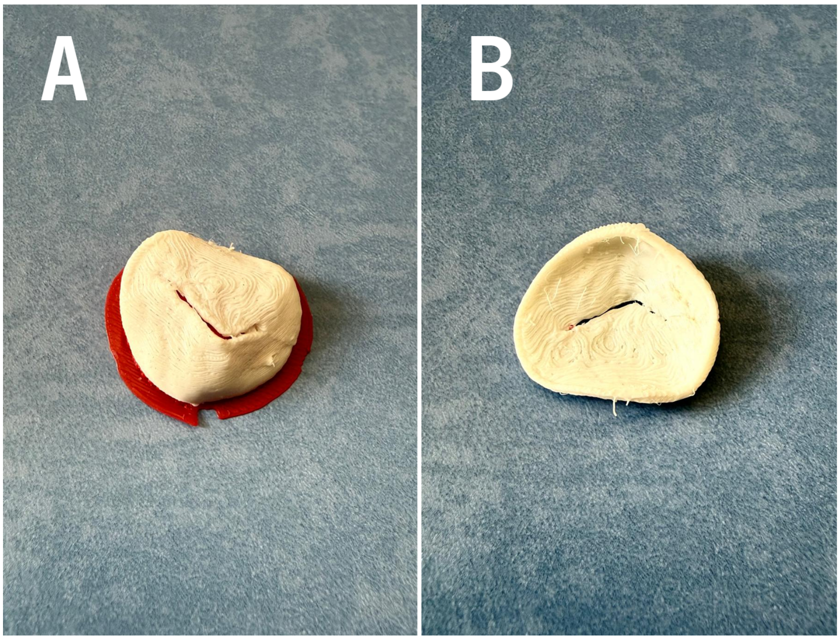


Figure 4.1: (A) top view of the completed valve prototype. (B) bottom view of the completed valve prototype. White material being TPU60A and red material being PLA support structures.

Figure 4.2 presents the photographs provided by Siemens Healthineers, displaying the valve appearing on the ultrasound images.

The feedback indicates that the manufactured valve can be visualized using ultrasound equipment. This is an important initial result, as ultrasound visibility represents a fundamental requirement for the intended application of the model as part of a physical ultrasound phantom. This shows that the valve can be used in larger phantom systems, i.e. in combination with the cardiac chamber phantom which is being developed in parallel by fellow student *Peter Machoritsch* at the Technical University of Vienna.

However, the assessment was qualitative. The clinical expert did deem it appropriate and plausible in appearance, but no quantitative measurements of acoustic properties, image contrast, signal intensity, or similarity to biological valve tissue were provided. Furthermore, the photographs document the appearance of the structure on the ultrasound display but do not establish the accuracy of its reconstructed thickness.

4.4 Evaluation of the Developed Pipeline

Considering the results of the reconstruction, manufacturing, and external evaluation, the developed pipeline achieved its principal technical objectives.

The initial volumetric ultrasound dataset and unthickened leaflet meshes were successfully

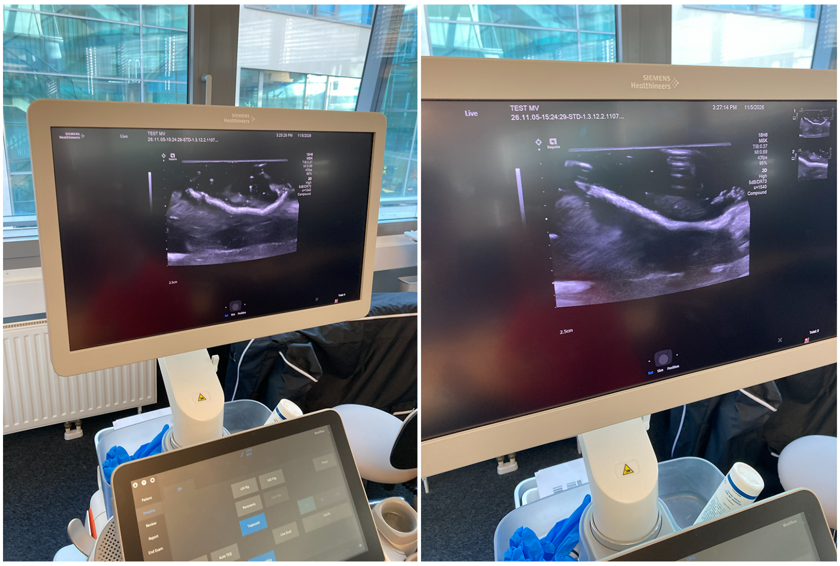


Figure 4.2: pictures of the valve appearing on ultrasound images taken by Dr. techn. Alexey Karimov while evaluating the prototype using the *ACUSON Sequoia Ultrasound System*

combined to reconstruct a solid mitral valve model with spatially varying thickness. The resulting geometry was suitable for additive manufacturing and could be fabricated using flexible TPU material.

Furthermore, the initial ultrasound assessment performed by Siemens Healthineers confirmed that the resulting physical prototype was visible using ultrasound imaging.

These findings demonstrate the feasibility of the proposed workflow, from volumetric ultrasound data to a manufactured valve prototype that can be visualized with ultrasound.

Nevertheless, the results should be interpreted within the scope of the conducted evaluation. The reconstructed thickness distribution was not independently compared against a physical reference measurement, and the ultrasound assessment did not include quantitative acoustic characterization. The prototype therefore represents a successfully manufactured and ultrasound-visible model, rather than a fully validated reproduction of the acoustic and mechanical properties of a biological accurate mitral valve.

4.5 Implications and Future Use-Cases

As the findings of this work demonstrate the feasibility of the developed pipeline, several potential future use-cases can be envisioned. One already touched upon is the integration of the manufactured valve into a larger ultrasound phantom system, which could be used for training and educational purposes. The ability to visualize the valve using ultrasound imaging suggests that it could serve as a valuable tool for medical professionals to practice and refine their skills in a controlled environment. Apart from that, procedural planning and device testing is already a well-established use-case for ultrasound phantoms [PCCSLSN26].

Actual implantation of 3D printed material in humans cardiovascular system still remains to this day as uncommon practice. There has been first successful research on the use of 3D printed stents but despite the promising results [FWL+24] there is a lot of progress to be made before a complex component like a heart valve can be manufactured via 3D printing and implanted in a human patient [LMR+20]. The main challenges are the mechanical properties of the printed material, the long-term durability of the implant, and the biocompatibility of the material. As long as the required biocompatibility, mechanical strength and longevity are not achieved, the clinical application of 3D printed heart valves will remain limited. However, as 3D printing technology continues to advance, it is possible that we may see more research and development in this area in the future. A benefit a patient-specific prosthesis could provide is the ability to tailor the design of the valve to the specific anatomy of the patient, potentially improving the fit and function of the implant. Additionally, 3D printing allows for rapid prototyping and customization, which could be particularly useful in cases where traditional manufacturing methods may not be feasible or cost-effective. Furthermore, the amount of foreign material placed inside the patient could be potentially minimized and therefore the risk of infection, thrombosis or prosthetic failure could be reduced [BYI+23].

CHAPTER 5

Conclusion

The objective of this thesis was to develop a reproducible pipeline for reconstructing spatially varying mitral valve leaflet thickness from volumetric ultrasound data and transferring the resulting geometry into a manufacturable physical model.

The developed approach combined volumetric ultrasound data with an initially unthickened surface mesh of the mitral valve leaflets. After establishing a common coordinate system, leaflet thickness was recovered through manually guided measurements in regions where the ultrasound boundaries could be identified reliably. This approach was selected over global automatic thresholding because of the considerable noise and segmentation artifacts present in the original dataset.

The resulting measurements were distributed across the leaflet surfaces using inverse distance weighting. Regions with insufficient image quality, particularly around the central opening and the outer leaflet boundaries, were treated using representative median thickness values. The interpolated thickness distributions were subsequently transferred to the original meshes and converted into closed solid geometries.

The resulting model was successfully manufactured using fused deposition modeling printing with flexible TPU60A filament. Despite several challenges associated with the printing of flexible materials and the removal of support structures, a satisfactory physical prototype was obtained. The final model retained the intended leaflet opening and exhibited a smooth underside following support structure removal.

An initial qualitative assessment was subsequently conducted by Siemens Healthineers. Feedback from the involved personnel confirmed that the manufactured valve was nicely scannable with ultrasound. The accompanying photographs of the ultrasound display provide additional qualitative evidence that the printed structure could be visualized using ultrasound equipment.

Returning to the research question, the results demonstrate that spatially varying leaflet thickness can be recovered from volumetric ultrasound data using a locally guided measurement strategy and transferred to a 3D-printable anatomical model. The successful manufacturing and subsequent ultrasound visualization demonstrate the practical feasibility of the proposed reconstruction pipeline.

However, several limitations remain. The thickness measurements depend on manual identification of leaflet boundaries and are therefore subject to operator-related uncertainty. The use of constant median values in regions with insufficient image quality introduces simplifications into the reconstructed thickness distribution. Furthermore, the interpolation provides an approximation of the thickness between measured locations rather than an independently verified representation of the true tissue thickness.

The ultrasound assessment also remains qualitative. Although the visibility of the prototype was confirmed, no quantitative comparison with biological tissue or an established reference phantom was performed. Consequently, the acoustic similarity and suitability of the prototype for more advanced ultrasound applications remain to be investigated.

Future work should focus on extending the evaluation to the complete heart phantom developed in parallel to this thesis project. In particular, integrating the mitral valve with the reconstructed heart chambers and submerging the complete model in an ultrasound medium would provide an opportunity to investigate ultrasound visualization under more representative experimental conditions.

In conclusion, this thesis establishes a practical workflow for reconstructing, manufacturing, and initially evaluating a patient-specific mitral valve prototype from volumetric ultrasound data. Although further validation is necessary before claiming equivalence to biological tissue, the results provide a promising foundation for the development of more realistic physical ultrasound phantoms of the cardiovascular system.

Appendix

Listing 5.1: Interpolation of the thickness distribution in Blender using inverse distance weighting.

```
import bpy
import numpy as np

def apply_thickness(obj_name, known_pos, known_val, power=2):
    obj = bpy.data.objects[obj_name]
    mesh = obj.data

    all_verts = np.array([v.co for v in mesh.vertices])

    def idw(query, kpos, kval, power):
        diff = query[:, None, :] - kpos[None, :, :]
        dist = np.maximum(np.linalg.norm(diff, axis=2), 1e-9)

        w = 1.0 / dist**power

        return (w * kval).sum(1) / w.sum(1)

    thickness_mm = idw(
        all_verts,
        np.array(known_pos),
        np.array(known_val),
        power
    )

    max_d = np.array(known_val).max()

    # Remove existing group for re-runs
    if "thickness" in obj.vertex_groups:
        obj.vertex_groups.remove(
            obj.vertex_groups["thickness"]
        )

    vg = obj.vertex_groups.new(name="thickness")

    for i, d in enumerate(thickness_mm):
        vg.add(
            [i],
            float(d / max_d),
            ’REPLACE’
        )

    print(
        f"{obj_name}: "
        f"{thickness_mm.min():.2f} - "
        f"{thickness_mm.max():.2f} mm "
        f"(max={max_d})"
    )

# ------------------------------------------------------------------
# Mesh 1
# ------------------------------------------------------------------
apply_thickness(
    "mesh1",
    known_pos=[
        [point1x, point1y, point1z],
        [point2x, point2y, point2z],
        # ... additional sample coordinates
        [pointN_x, pointN_y, pointN_z],
    ],
    known_val=[
        thickness1,
        thickness2,
        # ... additional thickness values
        thicknessN,
    ]
)

# ------------------------------------------------------------------
# Mesh 2
# ------------------------------------------------------------------
apply_thickness(
    "mesh2",
    known_pos=[
        [point1x, point1y, point1z],
        [point2x, point2y, point2z],
        # ... additional sample coordinates
        [pointM_x, pointM_y, pointM_z],
    ],
    known_val=[
        thickness1,
        thickness2,
        # ... additional thickness values
        thicknessM,
    ]
)
```

Table 5.1: Applied leaflet thickness and 3D spatial coordinates from the Blender model at all sampled locations, outer rim points, and center points for Mesh1 and Mesh2.

Point	Location	Leaflet	Coordinates (X, Y, Z) [mm]	Thickness [mm]
Mesh1				
1	Sample 1	Mesh1	(63.9590, 1.2938, -13.9180)	1.858
2	Sample 2	Mesh1	(64.5597, 0.8593, -11.7677)	1.695
3	Sample 3	Mesh1	(58.5578, -7.5698, -8.4560)	2.831
4	Sample 4	Mesh1	(65.7383, 16.3467, -7.1801)	1.493
5	Sample 5	Mesh1	(64.7657, 14.2849, -9.8768)	1.591
6	Sample 6	Mesh1	(65.3708, 26.0424, -4.8703)	2.703
7	Sample 7	Mesh1	(64.9545, 2.3063, -3.0277)	1.885
8	Sample 8	Mesh1	(66.2014, 7.9130, 0.3569)	1.418
9	Sample 9	Mesh1	(65.9559, 3.1092, 5.0442)	1.420
10	Sample 10	Mesh1	(62.5210, 8.6762, -13.1107)	1.660
11	Outer Rim 1	Mesh1	(60.0724, -5.3534, -13.0215)	2.634
12	Outer Rim 2	Mesh1	(60.8634, 2.2560, -16.6014)	2.634
13	Outer Rim 3	Mesh1	(59.9938, 8.3517, -15.4400)	2.634
14	Outer Rim 4	Mesh1	(60.5665, 15.1949, -13.5914)	2.634
15	Outer Rim 5	Mesh1	(62.8466, 20.5929, -11.7375)	2.634
16	Outer Rim 6	Mesh1	(65.2000, 25.0223, -9.6075)	2.634
17	Outer Rim 7	Mesh1	(65.8712, 27.4709, -6.3174)	2.634
18	Outer Rim 8	Mesh1	(64.6092, 29.0350, 0.8741)	2.634
19	Outer Rim 9	Mesh1	(59.9680, 11.1331, -14.7723)	2.634
20	Center 1	Mesh1	(63.6521, -0.7522, -0.0410)	1.420
21	Center 2	Mesh1	(66.8656, 9.9355, 4.1078)	1.420
22	Center 3	Mesh1	(66.5859, 16.3126, 2.0336)	1.420
23	Center 4	Mesh1	(66.5978, 20.5522, 0.9076)	1.420
Mesh2				
24	Sample 11	Mesh2	(55.8837, -7.7487, 3.9077)	2.391
25	Sample 12	Mesh2	(56.0970, -4.6908, 8.6596)	2.719
26	Sample 13	Mesh2	(59.9952, -3.0386, 6.9769)	1.444
27	Sample 14	Mesh2	(60.6347, -3.7250, 2.8164)	1.454
28	Sample 15	Mesh2	(63.1115, 3.0225, 11.4841)	2.013
29	Sample 16	Mesh2	(65.1467, 5.8269, 8.7319)	2.011
30	Sample 17	Mesh2	(60.3688, 16.5809, 15.3959)	2.634
31	Outer Rim 1	Mesh2	(63.7587, 28.0507, 2.7245)	2.634
32	Outer Rim 2	Mesh2	(63.4924, 25.6784, 7.3205)	2.634
33	Outer Rim 3	Mesh2	(61.9454, 21.5032, 11.7638)	2.634
34	Outer Rim 4	Mesh2	(60.0886, 15.0826, 16.3647)	2.634
35	Outer Rim 5	Mesh2	(58.0204, 8.4046, 18.1553)	2.634
36	Outer Rim 6	Mesh2	(56.3447, 3.7394, 17.3366)	2.634
37	Outer Rim 7	Mesh2	(55.1820, -0.8097, 13.2220)	2.634
38	Outer Rim 8	Mesh2	(55.8221, -6.6502, 5.8992)	2.634
39	Outer Rim 9	Mesh2	(56.0250, -8.6121, 1.7725)	2.634
40	Outer Rim 10	Mesh2	(56.9056, -8.9694, -3.2529)	2.634
41	Center 1	Mesh2	(63.7351, -1.3024, 0.4987)	1.420
42	Center 2	Mesh2	(64.8664, 1.7162, 5.0023)	1.420
43	Center 3	Mesh2	(66.8032, 5.1267, 6.6123)	1.420
44	Center 4	Mesh2	(66.8336, 9.7100, 4.8804)	1.420
45	Center 5	Mesh2	(66.7762, 14.3475, 3.3624)	1.420
46	Center 6	Mesh2	(66.1239, 20.7572, 1.0062)	1.420

Overview of Generative AI Tools Used

Generative AI tools were used to assist in the work conducted for this thesis as well of the writing of the thesis document itself. OpenAI's ChatGPT (GPT-6.1 Sol Medium) was used to generate initial drafts for sections of the thesis to set an initial direction for the discussed topics but the contents and wording and subsequent writing were performed by the author. AI was also used to generate a illustration of the pipeline schematic in Figure 3.1 as well as to assist in researching references for statements made in this thesis, which were checked by the author for suitability before inclusion. AI assistance was also provided for formatting and spelling correction of the document, however the actual phrasing and contents remained unaltered.

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